Stage 8 Maths — Practice Test 3

Paper 1: 60 minutes · no calculator · Paper 2: 60 minutes · calculator allowed · Total 100 marks

Answer all questions. Write your answers in the spaces provided. Show your working where there is space for it. The number of marks for each question is shown in brackets [ ].

Name _____ Class _____ Date _____ Mark: ____ / 100

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 60 minutes — 50 marks

1 **[1]**

(a) Work out $5 + 3 \times 2^2$

Working

Answer

(b) Work out $(-6) \times (-4) \div (-3)$ **[1]**

Working

Answer

2 **[1]**

Find the value of n.

$7^5 \times 7^3 \div 7^8 = 7^n$

Working

Answer

3 **[1]**

(a) Write 84 as a product of its prime factors.

(b) At a railway station, a train to Riverside leaves every 12 minutes and a train to Hilltop leaves every 18 minutes. **[2]**
Two trains, one to each town, leave together at 09:00.
At what time will a train to Riverside and a train to Hilltop next leave together? Show your working.

4 **[1]**

Which of these numbers is an integer but **not** a natural number?

- ☐ A) 7
- ☐ B) −3
- ☐ C) 0.5
- ☐ D) 2/3

5 **[1]**

(a) Work out $\sqrt[3]{(-64)}$

Working

Answer

(b) Youssef says, '−49 has two square roots: 7 and −7.' **[1]**
Is he correct? Explain your answer.

6 **[1]**

(a) Work out 46 × 0.01

Working

Answer

(b) Work out 0.75 ÷ 0.01 **[1]**

Working

Answer

7 **[2]**

(a) Work out 4 1/3 − 1 3/4
Give your answer as a mixed number in its simplest form. Show your working.

(b) Work out 6 × 2 2/3 **[1]**

Working

Answer

(c) Work out 5 ÷ 2/3 **[1]**

Working

Answer

(d) Which of these fractions is equal to a **recurring** decimal? **[1]**

- ☐ A) 3/8
- ☐ B) 7/20
- ☐ C) 5/6
- ☐ D) 9/25

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8 **[1]**

(a) Write the correct symbol, <, > or =, in the box.
−0.45 ☐ −4/9

(b) Work out 0.6 × 0.07 **[1]**

Working

Answer

(c) Work out 7.2 ÷ 0.3 **[1]**

Working

Answer

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9 **[2]**

(a) A youth club raises $1500 at a fundraising evening.
The money is shared between a charity, a club trip and new equipment in the ratio
charity : trip : equipment = 5 : 3 : 2
How much money is given to the club trip? Show your working.

Working

Answer $

(b) The club's banner is 40 cm wide and 2 m long. **[1]**
Write the ratio width : length in its simplest form.

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10 **[1]**

(a) Simplify $3a^2 + 5a - a^2 + 2a$

(b) Expand $2x(x^2 + 5)$ **[1]**

(c) Factorise completely $12y^2 + 18y$ **[2]**

(d) Which one of these is an **equation**? **[1]**

- ☐ A) $4n + 7$
- ☐ B) $P = 2l + 2w$
- ☐ C) $y > 2$
- ☐ D) $3t - 5 = 16$

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11 Solve these equations. **[2]**

(a) $5x - 7 = 2x + 8$
Show your working.

Working

Answer

(b) $x/4 + 3 = 1$ **[1]**

Working

Answer

12 The diagram shows an inequality on a number line. **[1]**



Write down the inequality shown, using x.

13 **[1]**

(a) The first term of a sequence is 3/4. The term-to-term rule is 'add 1/2'.
Work out the 4th term. Give your answer as a mixed number.

Working

Answer

(b) Here are the first four terms of another sequence. **[2]**
7, 4, 1, −2, …
Find an expression for the nth term.

14 Here is a function machine. **[1]**
input → [× 3] → [− 5] → output

(a) Find the output when the input is 4.

Working

Answer

(b) Find the input when the output is −3. **[1]**
Give your answer as a fraction.

Working

Answer

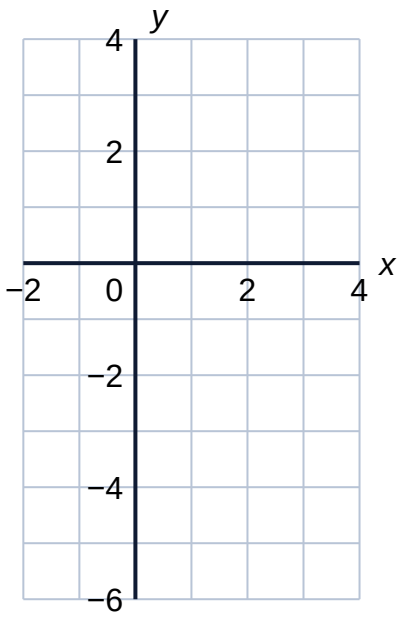
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15 **[1]**

(a) Which line has gradient −2 and crosses the y-axis at (0, 5)?

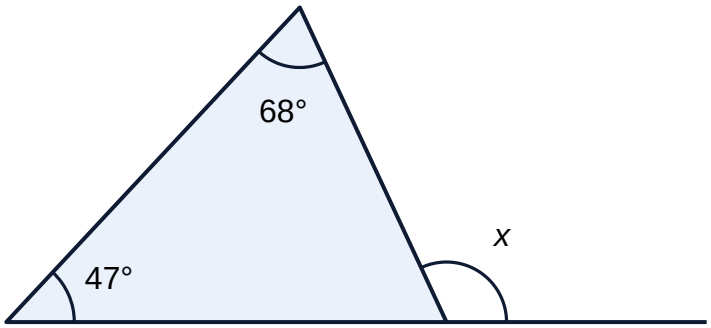
- ☐ A) y = −2x + 5
- ☐ B) y = 5x − 2
- ☐ C) y = 2x + 5
- ☐ D) y = −5x + 2

[2]



(b) Complete a table of values for y = 2x − 3 for x = −1, 0, 1, 2, 3.
Then draw the graph of y = 2x − 3 for values of x from −1 to 3 on a copy of this grid.
Draw the graph (on paper), then mark it with the mark scheme.

16 **[1]**



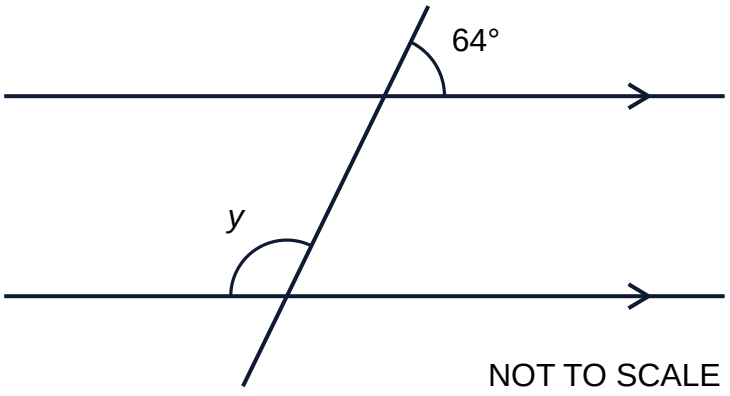
NOT TO SCALE

Work out the size of angle x.

Working

Answer ____ °

17 The two lines with arrows are parallel. **[1]**



NOT TO SCALE

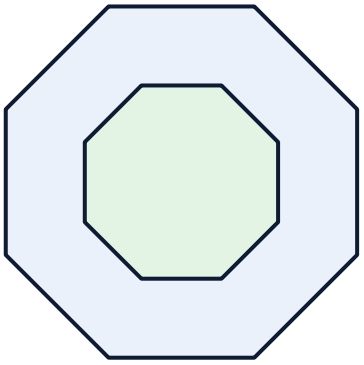
(a) Work out the size of angle y.

Working

Answer ____ °

(b) Give reasons for your answer to part (a). **[1]**

18 **[1]**



Plan view of the fountain

(a) A mosque courtyard has a fountain in its centre. From above, the fountain is a regular octagon.
Write down the number of lines of symmetry and the order of rotational symmetry of a regular octagon.

(b) The courtyard floor is decorated with quadrilaterals. **[1]**
Which statement is **always** true?

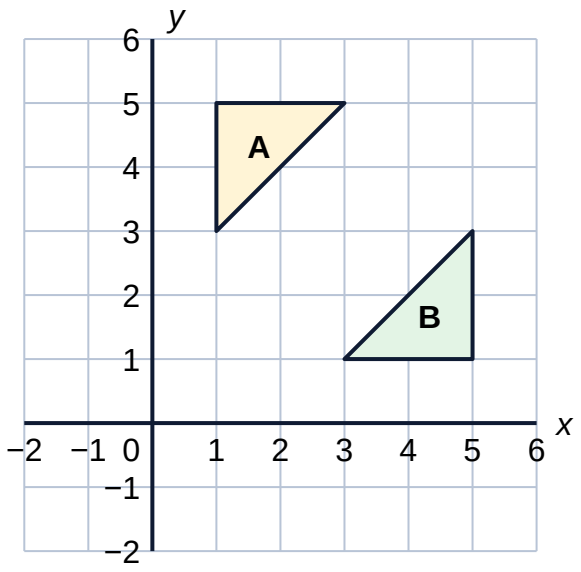
- ☐ A) A rhombus is a square.
- ☐ B) A trapezium is a parallelogram.
- ☐ C) A kite is a rhombus.
- ☐ D) A square is a rectangle.

(c) A lantern in the courtyard is a polyhedron with 12 vertices and 18 edges. **[1]**
Use Euler's formula to work out the number of faces of the lantern.

Working

Answer

19 **[2]**



Describe fully the **single** transformation that maps triangle A onto triangle B.

20 In a board game, a player spins a fair spinner numbered 1, 2, 3, 4 and rolls a fair six-sided dice numbered 1 to 6. **[1]**

(a) In this game, the probability that a player lands on a 'bonus' square on a turn is 0.18.
Work out the probability that the player does **not** land on a bonus square.

Working

Answer

(b) The player adds the number on the spinner and the number on the dice. **[2]**
Work out the probability that the total is 7. Show your working.

Working

Answer

Paper 2 — 60 minutes — 50 marks

1 An online shop is having a sale. **[2]**

(a) A jacket normally costs $64. In the sale its price is reduced by 15%.
Work out the sale price of the jacket, in dollars. Show your working.

Working

Answer

(b) After the sale, the price of a pair of headphones goes up from $48 to $54. **[2]**
Work out the percentage increase. Show your working.

Working

Answer %

(c) The price of a pencil case rises from $20 to $25. **[2]**
The price of a school bag rises from $80 to $92.
Hana says, 'The price of the school bag increased more.'
Explain why Hana is correct in one way but not correct in another way. Show your working.

2 **[2]**

(a) The online shop sells rice in two pack sizes.
Pack A: 750 g for $4.20
Pack B: 1.2 kg for $6.60
Which pack is better value for money? Show your working.

(b) 5 m of garden hose costs $13.75. The cost is directly proportional to the length. **[1]**
Work out the cost of 8 m of the same hose.

Working

Answer $

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3 A passenger train runs between several towns. **[1]**

(a) In one week the train travelled 6847.5 km.
Round 6847.5 to 2 significant figures.

Working

Answer

(b) The distance between two of the towns is 45 miles. **[1]**
Use 1 mile ≈ 1.6 km to change 45 miles into kilometres.

Working

Answer km

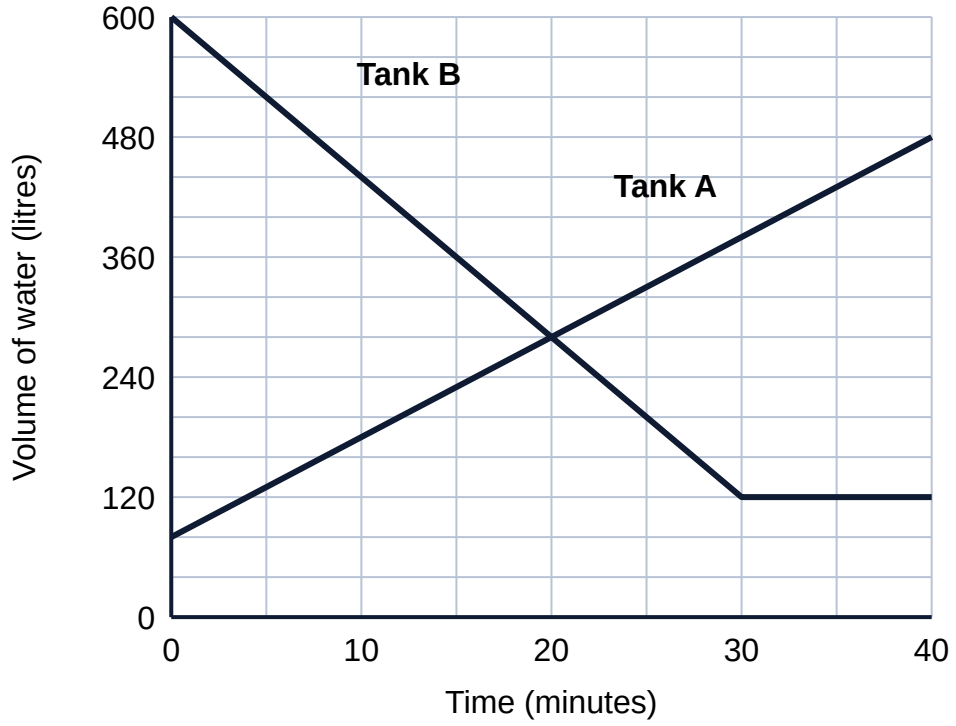
(c) The whole route is 120 km long. **[1]**
Use 1 km ≈ 5/8 mile to change 120 km into miles.

Working

Answer miles

4 Two water tanks, A and B, are connected to pipes. **[1]**

The graph shows the volume of water in each tank over 40 minutes.



(a) How many litres of water are in Tank B at the start?

Working

Answer ____ litres

(b) After how many minutes do the two tanks hold the same volume of water? **[1]**

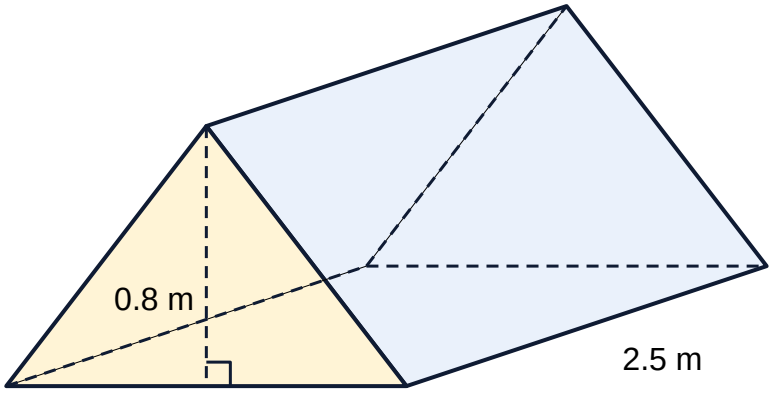
Working

Answer ____ minutes

(c) Describe what happens to the water in Tank B between 30 minutes and 40 minutes. Give a reason for the shape of the graph. **[1]**

(d) Tank A fills at a steady rate. Use the graph to find this rate, in litres per minute, and explain what the steady rate tells you about the shape of the graph for Tank A. **[1]**

5 **[2]**



NOT TO SCALE

(a) A water trough on a farm is a triangular prism.
Work out the volume of the trough. Show your working.

Working

Answer ____ m^3

(b) A water trough on a farm is a triangular prism. **[1]**
$1\ m^3 = 1000$ litres.
How many litres of water does the full trough hold?

Working

Answer ____ litres

(c) A closed water tank is a cuboid 2 m long, 1.5 m wide and 1.2 m high. **[2]**
The outside of the tank is painted. Work out its total surface area. Show your working.

Working

Answer ____ m^2

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6 A new courtyard is being designed for a mosque. **[2]**

(a) A circular pool in the courtyard has a diameter of 3.5 m.
Work out the circumference of the pool. Give your answer correct to 1 decimal place.

Working

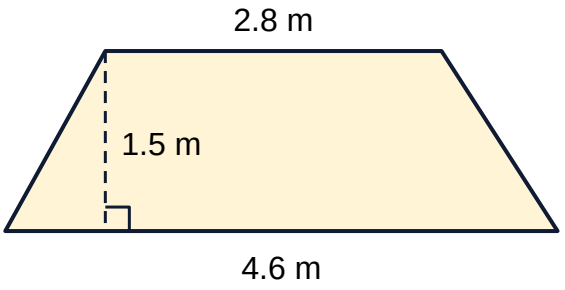
Answer ____ m

(b) A circular path around the courtyard has a circumference of 40 m. **[1]**
Work out the diameter of the path. Give your answer correct to 1 decimal place.

Working

Answer ____ m

[2]



NOT TO SCALE

(c) A marble panel on the courtyard wall is a trapezium.
Work out the area of the panel. Show your working.

Working

Answer ____ m^2

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7 The youth club held a car wash to raise money. The stem-and-leaf diagram shows the amounts, in dollars, collected by 15 boys. **[1]**

Stem	Leaf
1	2 5 8
2	0 3 3 7 9
3	1 4 4 6
4	0 2 8

Key: stem 2, leaf 3 means $23

(a) Find the median amount.

Working

Answer ____ $

(b) Find the range of the amounts. **[1]**

Working

Answer ____ $

(c) For 15 girls at the same car wash, the median amount was $33 and the range was $18. **[2]**
Make two comparisons between the amounts collected by the boys and the girls.

8 **[1]**

(a) Altogether, the youth club raised $2400 this year. It gave $1080 of this to charity.
The club draws a pie chart to show how the $2400 was used.
Work out the angle of the sector for charity.

Working

Answer °

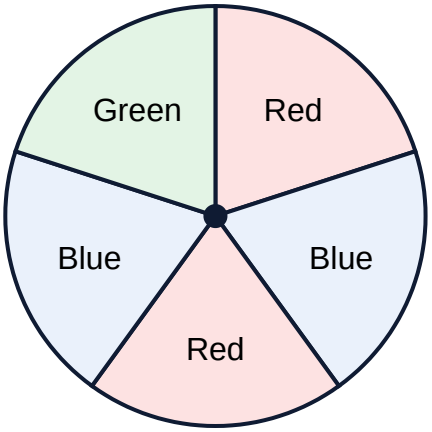
(b) The club has 240 members. The leader wants to know which activities the members would like next term. He decides to ask a sample of members. **[1]**
Which sample is **least** likely to be biased?

- ☐ A) Choose 30 members at random from the full list of 240 members
- ☐ B) Ask the first 20 members who arrive on Friday evening
- ☐ C) Ask 20 members of the club's football team
- ☐ D) Ask the leader's own friends in the club

(c) The club has 240 members. The leader wants to know which activities the members would like next term. He decides to ask a sample of members. **[1]**
Give one advantage and one disadvantage of asking a sample instead of asking all 240 members.

9 Karim makes a spinner for a board game. It has five equal sections: 2 red, 2 blue and 1 green. **[1]**
He spins it 200 times. The results are:

Colour	Red	Blue	Green
Frequency	74	85	41



(a) Work out the experimental probability of the spinner landing on red.

Working

Answer

(b) Write down the theoretical probability of the spinner landing on red. **[1]**

Working

Answer

(c) Karim says, 'My spinner must be biased, because 0.37 is not the same as my answer to part (b).' **[1]**
Is Karim correct? Explain your answer.

10 **[1]**

(a) At the youth club quiz night, an adult ticket costs $x and a child ticket costs $4 less than an adult ticket.
A family buys 2 adult tickets and 3 child tickets. They pay $43 altogether.
Show that 5x − 12 = 43.

(b) At the youth club quiz night, an adult ticket costs $x and a child ticket costs $4 less than an adult ticket. **[1]**
A family buys 2 adult tickets and 3 child tickets. They pay $43 altogether.
Solve the equation to find the cost of an adult ticket.

Working

Answer $

(c) The online shop uses this formula for the delivery cost, C dollars, of a parcel with a mass of w kilograms: **[1]**
C = 4.5 + 0.8w
Work out the delivery cost for a parcel with a mass of 12.5 kg.

Working

Answer $

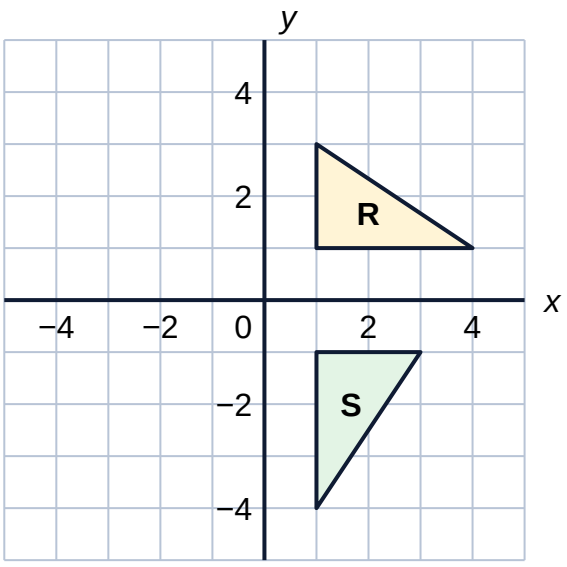
(d) The online shop uses this formula for the delivery cost, C dollars, of a parcel with a mass of w kilograms: **[1]**
C = 4.5 + 0.8w
Rearrange the formula to make w the subject.

(e) The youth club has p members. Two-thirds of the members go on a trip, and 5 leaders go with them. **[1]**
Write an expression, in terms of p, for the total number of people who go on the trip.

11 **[1]**

(a) Triangle T has vertices at (2, 3), (3, 3) and (2, 5).
T is enlarged by scale factor 3, with centre of enlargement (1, 1).
Write down the coordinates of the image of the vertex (2, 5).

[2]



(b) Describe fully the single transformation that maps triangle R onto triangle S.

12 **[1]**

(a) On a plan of the mosque courtyard, the bearing of the minaret from the gate is 125°.
Work out the bearing of the gate from the minaret.

Working

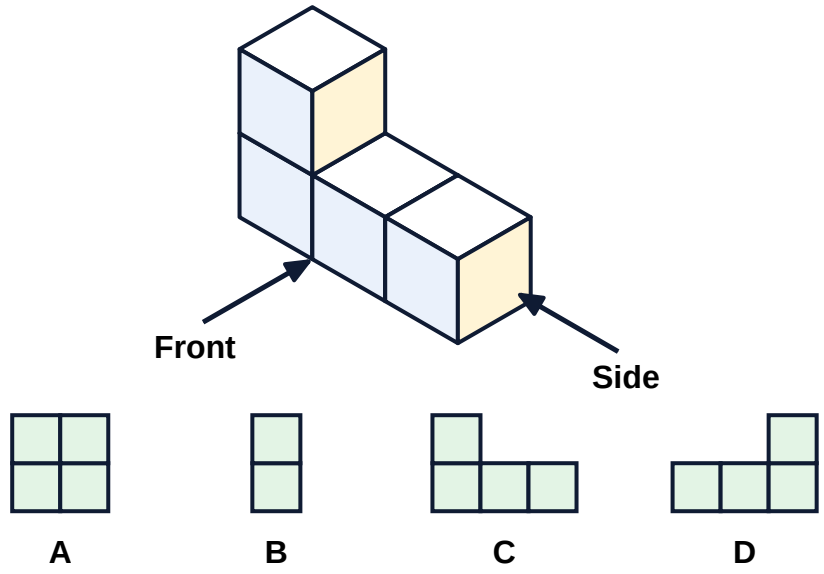
Answer °

(b) On the same plan, a straight path goes from (−3, 5) to (7, −1). A lamp is placed at the midpoint of the path. **[1]**
Find the coordinates of the lamp.

13 A triangular flower bed in the courtyard is drawn to scale as triangle ABC with AB = 7 cm, AC = 5 cm and BC = 6 cm. **[2]**

On paper, use a ruler and compasses only to construct triangle ABC. Leave in your construction arcs.

14 The diagram shows a shape made from four cubes, and four drawings A, B, C and D. **[1]**



(a) Which drawing is the **front elevation** of the shape (the view in the direction of the arrow marked Front)?

- ☐ A) A
- ☐ B) B
- ☐ C) C
- ☐ D) D

(b) On squared paper, draw the plan (the view from above) of the shape. **[1]**